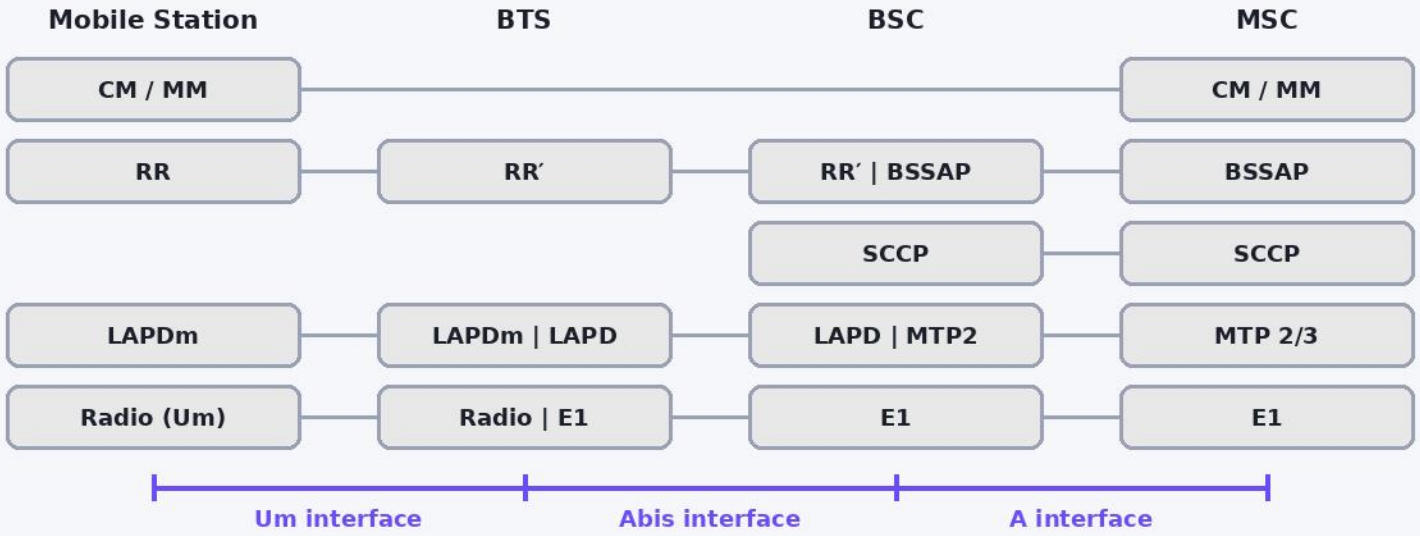
Ex-4 DESIGN: Mobile network protocol stack – layout and elements



Design specifications

Frame: Desktop 1440 x 800

Columns: MS, BTS, BSC, MSC

Rows: 5 layers (top to bottom)

Cell: Rectangle, radius 10

Peer link: 3 px line

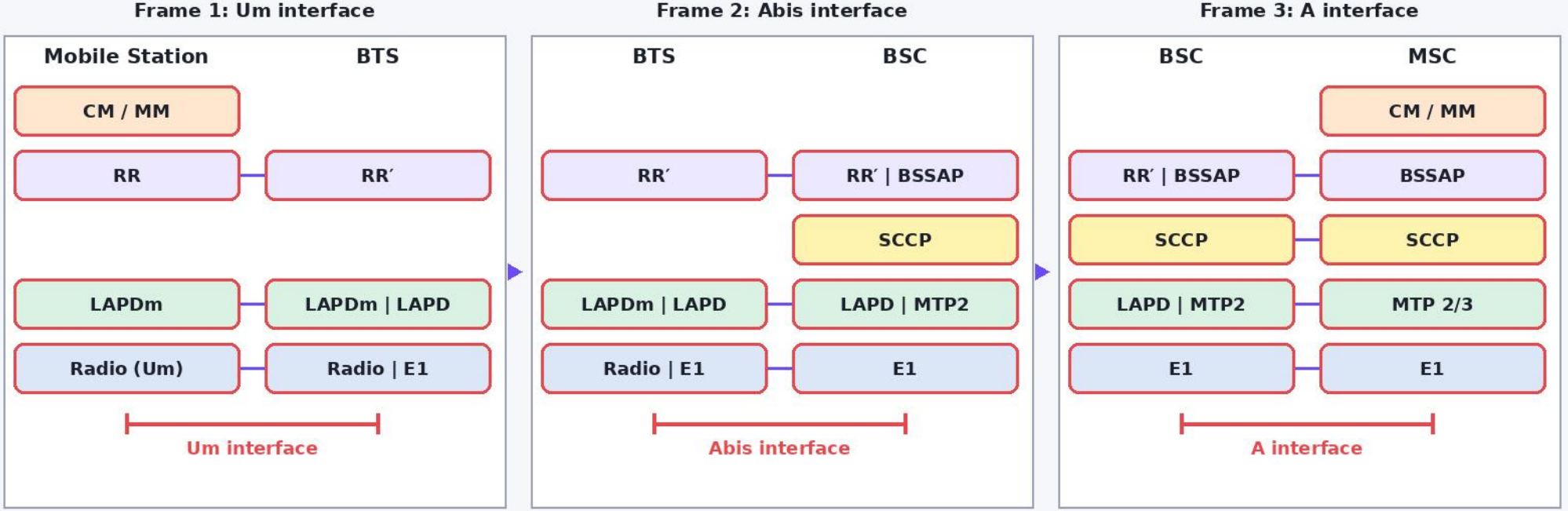
Interface bar: Um, Abis, A

Font: Inter Bold 18

Layer colours:

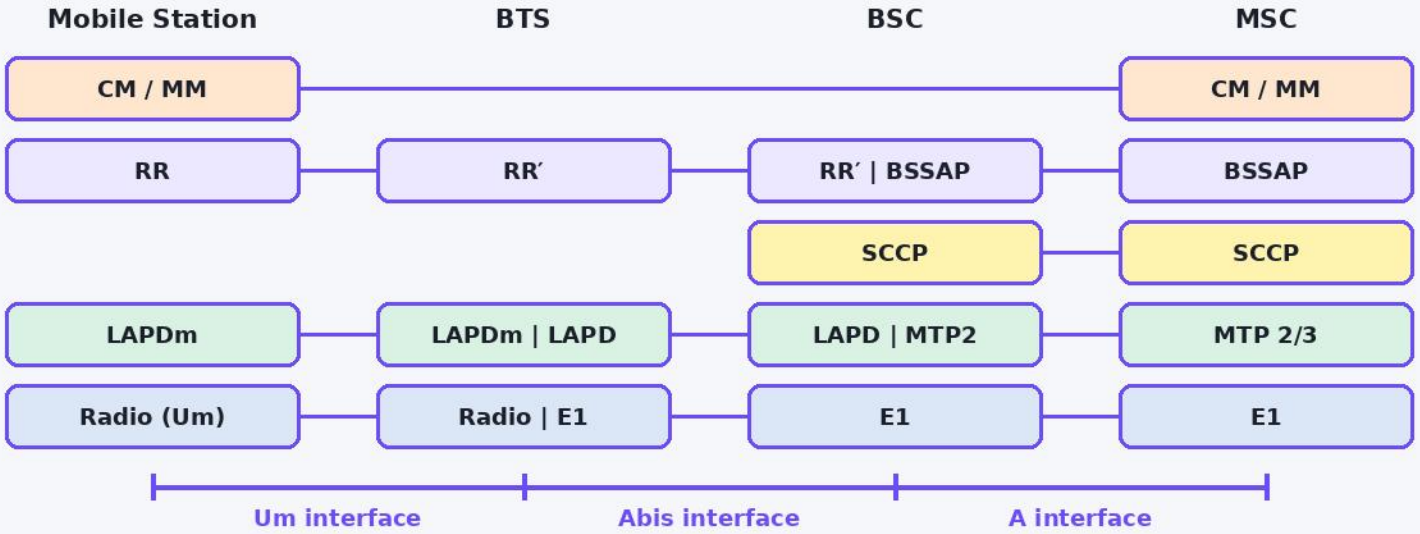
- Application orange
- Signalling lilac
- Transport yellow
- Data link green
- Physical blue

Ex-4 PROTOTYPE: Click an interface to highlight its protocol stack (Navigate to, Smart Animate)



Trigger: On click interface tab | Action: Navigate to | Animation: Smart animate, 300 ms | Selected frame outlined red

Ex-4 OUTPUT: Key elements of the mobile network protocol stack (GSM signalling)



Key elements

- CM: call control
- MM: mobility, location
- RR: radio resources
- BSSAP: BSC–MSC messages
- SCCP / MTP: SS7 transport
- LAPDm / LAPD: link layer
- Um: radio air interface
- Abis: BTS–BSC link
- A: BSC–MSC link

Application Signalling Transport Data link Physical